VECTORS AND SCALARS AND COMPONENTS OF A VECTOR

All physical quantities can be grouped into scalars and vectors. In Physics, scalars and vectors are used to explain concepts. Mass, distance and speed are examples of scalars. Displacement, velocity, acceleration and force are examples of vectors. Also, this session teaches you how to break a given vector into its components. A vector lying in an x-y plane and directed at an angle has both x-component and y-component. How to determine magnitude and direction of vectors are also discussed. This session introduces you to another method of adding two vectors. The parallelogram law of vectors is used to find the resultants of two vectors. This law can be applied graphically or by calculation to find the resultant of two vectors.

Objectives

At the end of this session you should be able to:

1. explain scalars and give examples.
2. explain vectors and give examples
3. distinguish between a scalar and a vector.
4. components of any given vector.
5. magnitudes of vectors.
6. directions of vectors.
7. define a unit vector.
8. find the unit vector of a given vector.
9. write a vector in the position vector notation.
10. add and subtract vectors graphically.
11. add and subtract vectors by calculations.
12. apply the laws of vector algebra to do calculations.
13. understand the parallelogram law of vectors;
14. to use the law to find resultants of two or more vectors.

2.1 Vectors and Scalars

In Physics, physical quantities are grouped into scalars and vectors. Not all physical quantities involve a direction.

A scalar quantity is specified by a single value with an appropriate unit and has no direction. That is, a scalar quantity has only magnitude but no direction. Physical quantities like temperature, volume, mass, time intervals, pressure and energy are examples of scalars because they have only magnitudes with unit.

A vector quantity has magnitude as well as direction. It is typically represented by an arrow whose direction is the same as that of the quantity and whose length is proportional to the quantity's magnitude. The direction of a vector is often expressed as an anticlockwise angle of rotation of the vector about its "tail" (starting point) from due East. Using this convention, a vector **E** with a direction of 40 degrees is a vector that has been rotated 40 degrees in an anticlockwise direction relative to due east as shown in Figure 1.1. Examples of vector quantities are displacement, velocity and acceleration.

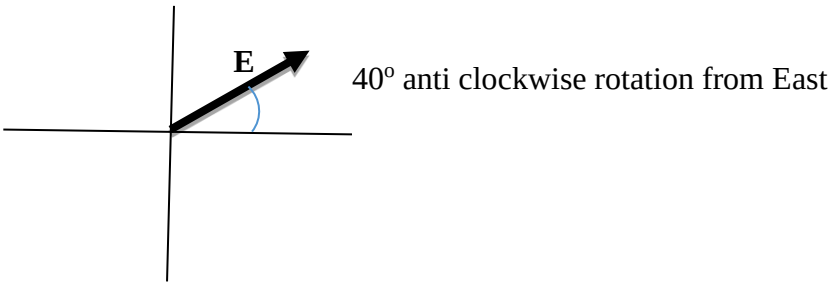


Figure 1.1 Vector **E** is rotated through an angle 40 degrees in the anticlockwise direction.

2.2 Components of a Vector

A vector that starts from the origin and is directed along the customary axes (ie. x, y, and z) will have only one component. For example, if a vector **D** moves 2 units along the x-axis, its components will read as **D** (2,0) which means that it has only x-component.

In situations where a vector is directed at angles to the customary coordinate axes, a useful method will be employed to transform the vector into two parts with each part being directed along the coordinate axes. For example, a vector that is directed northwest can be thought of as having two parts (ie. a northward part and a westward part). A vector with coordinate **B** (3,4) will have 3 units along the x-axis and 4 units along the y-axis.

Consider a vector **A** lying in the *xy* plane and making an arbitrary angle θ with the positive *x* axis as shown in Figure 1.2.

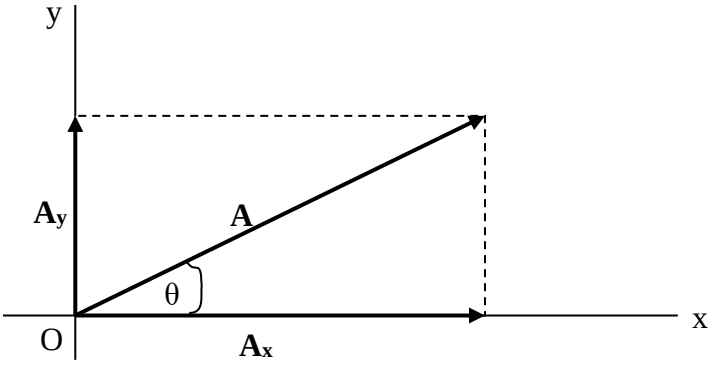


Figure 1.2 Any vector **A** lying in the xy plane can be represented by a vector $\mathbf{A}_x$ lying along the x axis and by a vector $\mathbf{A}_y$ lying along the y axis, where $\mathbf{A} = \mathbf{A}_x + \mathbf{A}_y$

UNIT VECTORS AND LAWS OF VECTOR ALGEBRA

In this session you will be introduced to unit vectors which are used to specify a given direction and have no other physical significance so that it lacks both dimension and unit. They help you to write vectors in vector notation. Unit vectors also help you to identify the components of vectors with ease.

2.3 Unit Vectors

A unit vector is a vector that has a magnitude of 1 and points in a particular direction. It is denoted by $\hat{u}$. Unit vectors are used to specify a given direction and have no other physical significance so that it lacks both dimension and unit. The unit vectors in the positive directions of the *x, y, z* axes are represented by $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$ as shown in Figure 1.3. The unit vectors form a set of mutually perpendicular vectors in a right-handed coordinate system as shown in Figure 1.3.
Mathematically, a unit vector is given by the ratio of a vector to its magnitude.

That is $\hat{u} = \dfrac{\bar{u}}{|u|}$.

The magnitude of each unit vector equals 1; that is $|\mathbf{i}| = |\mathbf{j}| = |\mathbf{k}| = 1$.

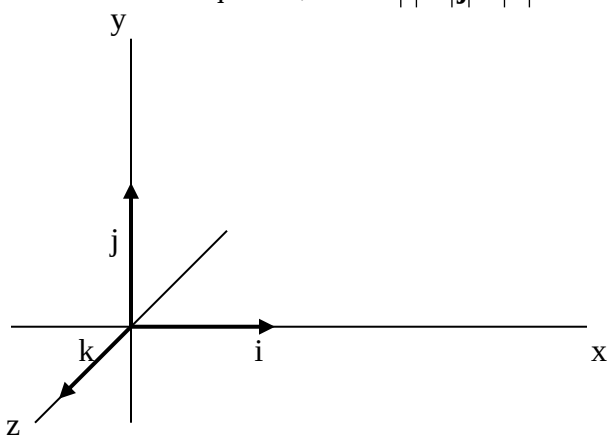


Figure 1.3 Unit vectors **i**, **j** and **k** define the directions of a right-handed rectangular coordinate system.

Consider a vector **A** lying in the *xy* plane as shown in Figure 1.4. The product of the component A_x and the unit vector **i** is the vector $A_x\mathbf{i}$, which lies on the *x* axis and has magnitude $|A_x|$. $A_y\mathbf{j}$ is a vector of magnitude $|A_y|$ lying on the *y* axis.

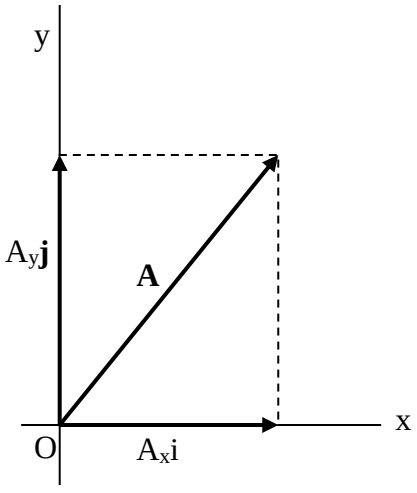


Figure 1.4 Vector $A = A_x\mathbf{i} + A_y\mathbf{j}$ lying in the xy plane has components A_x and A_y

The unit vector notation for the vector **A** then is

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} \qquad \textbf{1.5}$$

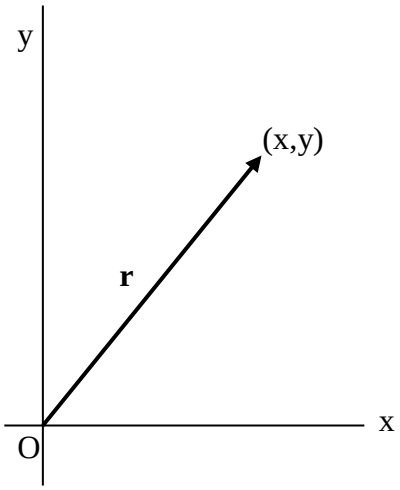


Figure 1.5 The point whose Cartesian coordinates are (x, y) can be represented by the position vector r = x**i** + y**j**

If a point lies in the xy plane with Cartesian coordinates (x, y) as shown in Figure 1.5**,** then its **position vector** $\mathbf{r}$ in unit-vector form is given by

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} \qquad \textbf{1.6}$$

Equation 1.6 tells us that the components of $\mathbf{r}$ are the lengths x and y .

If we want to add vectors **A** and **B** in component form so that the resultant is $\mathbf{R} = \mathbf{A} + \mathbf{B}$ then,

$$\mathbf{R} = \left(A_x\mathbf{i} + A_y\mathbf{j}\right) + \left(B_x\mathbf{i} + B_y\mathbf{j}\right) \qquad \textbf{1.7}$$

or

$$\mathbf{R} = \left(A + B\right)\mathbf{i} + \left(A + B\right)\mathbf{j} \qquad \textbf{1.8}$$

But $\mathbf{R} = R_x\mathbf{i} + R_y\mathbf{j}$, so that the components of the resultant vector are

$$\begin{aligned} R_x &= A_x + B_x \\ R_y &= A_y + B_y \end{aligned} \qquad \textbf{1.9}$$

The magnitude of **R** then becomes

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{\left(A_x + B_x\right)^2 + \left(A_y + B_y\right)^2} \qquad \textbf{1.10}$$

and the angle that **R** makes with the x axis is

$$\tan\theta = \frac{R_y}{R_x} = \frac{A_y + B_y}{A_x + B_x} \qquad \textbf{1.11}$$

If **A** and **B** both have x, y and z components, they can be expressed in the form

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$$
$$\mathbf{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k} \qquad \textbf{1.12}$$

The sum of **A** and **B** is

$$\mathbf{R} = \left(A_x + B_x\right)\mathbf{i} + \left(A_y + B_y\right)\mathbf{j} + \left(A_z + B_z\right)\mathbf{k} \qquad \textbf{1.13}$$

2.4 Laws of Vector Algebra

If **A**, **B** and **C** are vectors and p and q are scalars, then

1. $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ Commutative Law for Addition
2. $\mathbf{A} + (\mathbf{B} + C) = (\mathbf{A} + B) + C$ Associative Law for Addition
3. $p(q\mathbf{A}) = (pq)\mathbf{A} = q(p\mathbf{A})$ Associative Law of Multiplication
4. $(p + q)\mathbf{A} = p\mathbf{A} + q\mathbf{A}$ Distributive Law
5. $p(\mathbf{A} + \mathbf{B}) = p\mathbf{A} + p\mathbf{B}$ Distributive Law

Note that in these Laws only multiplication of a vector by one or more scalars is defined.

Example 1.2. A 5N force acts on a body at an angle of 30º to the horizontal.

i. Find the components of the Force F.
ii. Write down F in the position vector notation
iii. Find the unit vector of F

Solution

i. Using the trigonometric ratios,

The force along the horizontal direction $F_x = F\cos\theta = 5\cos30^o = 4.33N;$

The force along the vertical direction $F_y = F\sin\theta = 5\sin30^o = 2.5N$

ii. $F = 4.33\boldsymbol{i} + 2.5\boldsymbol{j}$

iii. Unit vector $\hat{F} = \frac{\vec{F}}{|F|} = \frac{4.33\mathrm{i} + 2.5\mathrm{j}}{5} = \frac{4.33}{5}\boldsymbol{i} + \frac{2.5}{5}\boldsymbol{j}$

Example 1.3. Given a vector A = 2i + 5j, find the

i. magnitude of vector A
ii. direction of the vector A

Solution

i. Magnitude of vector $|A| = \sqrt{2^2 + 5^2} = \sqrt{4 + 25} = 5.4\ unit$

ii. $tan\theta = \frac{5}{2} = 2.5$

$\theta = \tan^{-1} 2.5 = 68.2^0$

Example 1.4. A particle traveled a distance of 10 km from point O to point A at an angle of 60° north-east, then another 15 km to point B due east. Find the

i. position vector $\overrightarrow{OB}$
ii. distance between points O and B?
iii. direction of vector $\overrightarrow{OB}$

Solution

i. $\overrightarrow{OA} = 10\ cos60^0\boldsymbol{i} + 10\ sin60^0\boldsymbol{j} = 5\boldsymbol{i} + 8.66\boldsymbol{j}$

$\overrightarrow{AB} = 15\ cos0^0\boldsymbol{i} + 10\ sin0^0\boldsymbol{j} = 15\boldsymbol{i} + 0\boldsymbol{j}$

$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = (5 + 15)\boldsymbol{i} + (8.66 + 0)\boldsymbol{j} = 20\boldsymbol{i} + 8.66\boldsymbol{j}$

ii. The distance between points O and B is given by

$|\overrightarrow{OB}| = \sqrt{20^2 + 8.66^2} = 21.8\ m$

iii. $tan\theta = \frac{8.66}{20} = 0.433$

$\theta = \tan^{-1} 0.433 = 23.4^0$

Example 1.5. An expeditionary team in search of antiques in a cave travels a net 2.6km westward, 3.9km southward, and 25m upward. What was their displacement vector?

Solution

Let their horizontal displacement be d_h. Then by the Pythagoras theorem the magnitude of d_h is

$$d_h = \sqrt{(2.6km)^2 + (3.9km)^2} = 4.69km$$

The angle θ of their displacement vector relative to the west is given by

$$\tan\theta = \frac{3.9km}{2.6km} = 1.5$$

or

$$\theta = \tan^{-1} 1.5 = 56^0$$

We now find the magnitude of the overall displacement d,

$d = \sqrt{(4.69km)^2 + (0.025km)^2} = 4.69km$ and the angle ϕ,

$$\phi = \tan^{-1} \frac{0.025km}{4.69km} = 0.3^0 .$$

Example 1.6. Find the sum of two vectors **A** and **B** lying in the *xy* plane and given by $\mathbf{A} = (2\mathbf{i} + 2\mathbf{j})\text{m}$ and $\mathbf{B} = (2\mathbf{i} - 4\mathbf{j})\text{m}$.

Solution

The general expression for **A** is $\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j}$ and comparing we see that A_x = 2m and A_y = 2m. Also, B_x = 2m and B_y = -4m. Using Equation 1.8 to obtain the resultant vector **R** we have

$$\mathbf{R} = \mathbf{A} + \mathbf{B} = (2+2)\mathbf{i}\,m + (2-4)\mathbf{j}\,m$$

$$= (4\mathbf{i} - 2\mathbf{j})m$$

or

$R_x = 4m$ and $R_y = -2m$

The magnitude of **R** is given by Equation 2.10:

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{(4m)^2 + (-2m)^2} = 4.5m$$

From Equation 1.11 we find the direction of **R** as,

$$\tan\theta = \frac{R_y}{R_x} = \frac{-2m}{4m} = -0.5$$

$\theta = \tan^{-1}(-0.5) = -27^0$ clockwise from the x-axis.

Example 1.7. A car travels 20km due north and then 35km in a direction 60^0 west of north, as shown in Figure 1.6. Find the magnitude and direction of the car's resultant displacement.

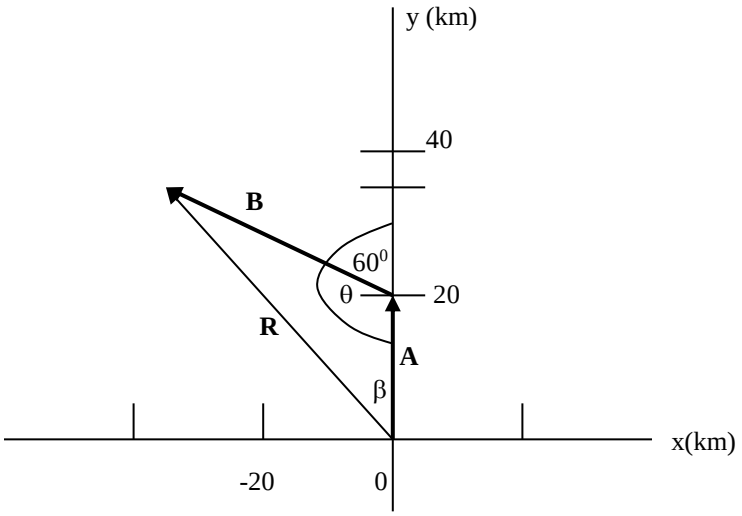


Figure 1.6 Graphical method for finding the resultant displacement vector **R** = **A** + **B**

Solution

This problem can be solved in two different ways: 1. Geometrical method using graph paper and protractor and by calculation. The geometrical method is as in Figure 1.6. The displacement **R** is the resultant when the two individual displacements **A** and **B** are added.

Algebraically, the magnitude of the resultant **R** can be obtained from the law of cosines as applied to the triangle. The angle is given as $\theta = 180^0 - 60^0 = 120^0$ and $R^2 = A^2 + B^2 - 2AB\cos\theta$, so that

$$R = \sqrt{A^2 + B^2 - 2AB\cos\theta}$$

$$R = \sqrt{(20km)^2 + (35km)^2 - 2(20km)(35km)\cos 120^0}$$

$$R = 48.2km$$

From the law of sines, the direction of **R** is

$$\frac{\sin\beta}{B} = \frac{\sin\theta}{R}$$

$$\sin\beta = \frac{B}{R}\sin\theta = \frac{35km}{48.2km}\sin 120^0 = 0.629$$

$$\beta = 38.9^0$$

The resultant displacement of the car is $48.2km$ in a direction 38.9^0 west of north.

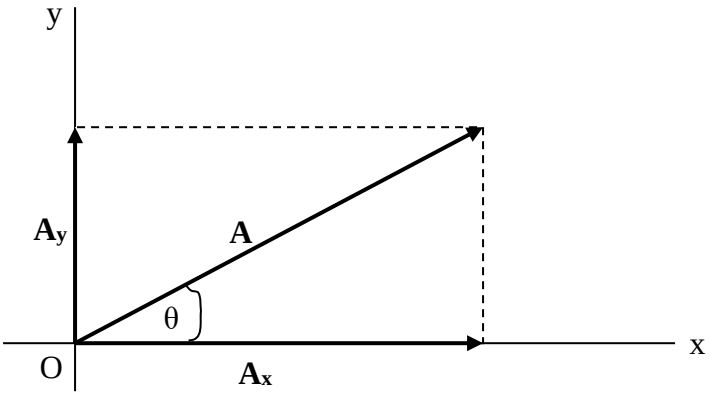


Figure 1.2 Any vector **A** lying in the xy plane can be represented by a vector $\mathbf{A_x}$ lying along the x axis and by a vector $\mathbf{A_y}$ lying along the y axis, where $\mathbf{A} = \mathbf{A_x} + \mathbf{A_y}$

This vector can be expressed as the sum of two other vectors $\mathbf{A}_x$ and $\mathbf{A}_y$. From Figure 1.2, the three vectors form a right triangle and that $\mathbf{A} = \mathbf{A}_x + \mathbf{A}_y$. The vectors $\mathbf{A}_x$ and $\mathbf{A}_y$ are often referred to as "the components of the vector **A**," written as A_x and A_y (with no boldface). The component A_x represents the projection of **A** along the *x* axis, and the component A_y represents the projection of **A** along the *y* axis. If $\mathbf{A}_x$ points in the positive *x* direction the component A_x is positive and *vice versa*. The same is true for the component A_y. From Figure 1.2 and the definition of sine and cosine, then $\cos\theta = A_x/A$ and $\sin\theta = A_y/A$. Hence the components of **A** are

$$A_x = A\cos\theta \qquad \textbf{1.1}$$

$$A_y = A\sin\theta \qquad \textbf{1.2}$$

It follows from Figure 1.1 that the magnitude and direction of **A** are related to its components through the expressions

$$A = \sqrt{A_x^2 + A_y^2} \qquad \textbf{1.3}$$

$$\theta = \tan^{-1}\left(\frac{A_x}{A_y}\right) \qquad \textbf{1.4}$$

Notice that the signs of the components A_x and A_y depend on the angle θ. As an example, if $\theta = 125^0$, then A_x is negative and A_y is positive. If $\theta = 235^0$, then both A_x and A_y are negative.

GRAPHICAL ADDITION AND SUBTRACTION OF VECTORS

In this session you are taught how to add or subtract any number of given vectors graphically and also by calculation. For graphical work you will need a ruler, protractor, a sheet of paper and a pencil. You are also introduced to the Laws of vector algebra to guide you in your vector calculations.

2.5 Vector Algebra

A vector is represented analytically by a bold-faced letter such as **A.** Geometrically it is represented by an arrow PQ where P is called the initial point and Q is called the terminal point. The magnitude or length of the vector is then denoted by $|A|$.

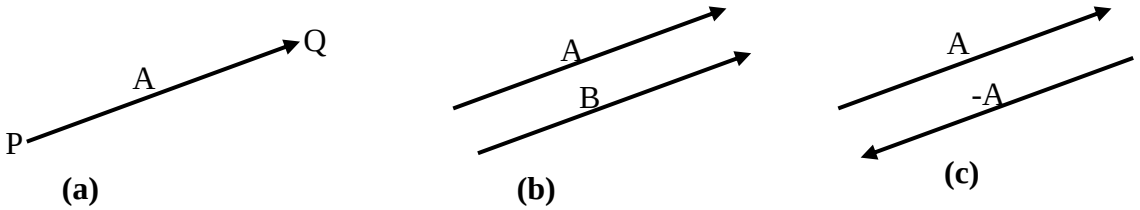


Figure 1.7: (a) A position vector **A** between two points P and Q. **(b)** Two vectors **A** and **B** having same magnitude and same direction (c) The vectors **A** and **-A** have the same magnitude but are in opposite direction.

The operations of addition, subtraction and multiplication familiar in the algebra of real numbers are with suitable definitions capable of extension to an algebra of vectors.
The following definitions are fundamental.

1. Two vectors **A** and **B** are equal if they have the same magnitude and direction regardless of their initial points. Thus **A** = **B** in figure 1.7b.
2. A vector having direction opposite to that of vector **A** but with the same length is denoted by –**A** as in figure 1.7c.
3. The sum or resultant of vectors **A** and **B** of figure 1.8a is a vector **C** formed by placing the initial point of **B** on the terminal point of **A** and joining the initial point of **A** to the terminal point of **B** [see figure 1.8b]. We write **C** = **A** + **B**. This definition is equivalent to the parallelogram law for vector addition as indicated in figure 1.8c.

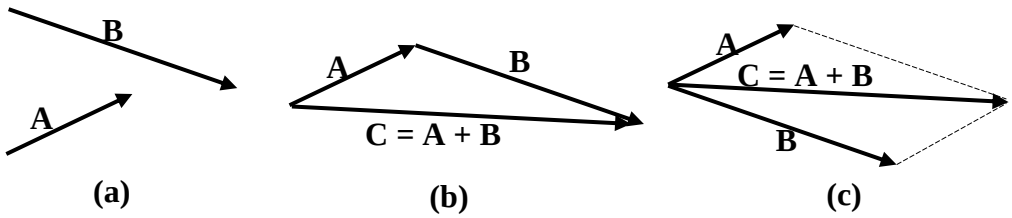


Figure 1.8: (a) A and **B** are two independent vectors. **(b)** Vector **C** is the sum or resultant of the vectors **A** and **B** (c) Vector **C** is the sum or resultant of the vectors **A** and **B** showing the parallelogram law of vector addition.

1. The difference of vectors **A** and **B**, represented by **A** – **B**, is that vector **C** which when added to **B** gives **A**. Equivalently, **A** – **B** may be defined as **A** + (**- B**). If **A** = **B**, then **A** – **B** is defined as the null or zero vector represented by 0. This has a magnitude of zero but its direction is not defined.
2. The product of a vector **A** by a scalar ρ is a vector ρ**A** or **A**ρ with magnitude $|\rho|$ times the magnitude of **A** and direction the same as or opposite to that of **A** according as ρ is positive or negative. If ρ = 0, ρ**A** = 0, the null vector.

DOT OR SCALAR PRODUCT

The dot or scalar product of two vectors **A** and **B**, denoted by **A**·**B** (read **A** dot **B**) is defined as the product of the magnitudes of **A** and **B** and the cosine of the angle between them. In symbols,

$$\mathbf{A}\cdot\mathbf{B} = AB\cos\theta, \qquad 0 \leqq \theta \leqq \pi$$

Note that **A**·**B** is a scalar and not a vector.

The following laws are valid:

1. $\mathbf{A}\cdot\mathbf{B} = \mathbf{B}\cdot\mathbf{A}$ Commutative Law for Dot Products
2. $\mathbf{A}\cdot(\mathbf{B}+\mathbf{C}) = \mathbf{A}\cdot\mathbf{B} + \mathbf{A}\cdot\mathbf{C}$ Distributive Law
3. $p(\mathbf{A}\cdot\mathbf{B}) = (p\mathbf{A})\cdot\mathbf{B} = \mathbf{A}\cdot(p\mathbf{B}) = (\mathbf{A}\cdot\mathbf{B})p$, where p is a scalar.
4. $\mathbf{i}\cdot\mathbf{i} = \mathbf{j}\cdot\mathbf{j} = \mathbf{k}\cdot\mathbf{k} = 1, \quad \mathbf{i}\cdot\mathbf{j} = \mathbf{j}\cdot\mathbf{k} = \mathbf{k}\cdot\mathbf{i} = 0$
5. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, then
$$\mathbf{A}\cdot\mathbf{B} = A_1B_1 + A_2B_2 + A_3B_3$$
$$\mathbf{A}\cdot\mathbf{A} = A^2 = A_1^2 + A_2^2 + A_3^2$$
$$\mathbf{B}\cdot\mathbf{B} = B^2 = B_1^2 + B_2^2 + B_3^2$$
6. If $\mathbf{A}\cdot\mathbf{B} = 0$ and **A** and **B** are not null vectors, then **A** and **B** are perpendicular.

Evaluate each of the following.

(a) $\mathbf{i}\cdot\mathbf{i} = |\mathbf{i}|\,|\mathbf{i}|\cos 0° = (1)(1)(1) = 1$

(b) $\mathbf{i}\cdot\mathbf{k} = |\mathbf{i}|\,|\mathbf{k}|\cos 90° = (1)(1)(0) = 0$

(c) $\mathbf{k}\cdot\mathbf{j} = |\mathbf{k}|\,|\mathbf{j}|\cos 90° = (1)(1)(0) = 0$

(d) $\mathbf{j}\cdot(2\mathbf{i} - 3\mathbf{j} + \mathbf{k}) = 2\mathbf{j}\cdot\mathbf{i} - 3\mathbf{j}\cdot\mathbf{j} + \mathbf{j}\cdot\mathbf{k} = 0 - 3 + 0 = -3$

(e) $(2\mathbf{i} - \mathbf{j})\cdot(3\mathbf{i} + \mathbf{k}) = 2\mathbf{i}\cdot(3\mathbf{i} + \mathbf{k}) - \mathbf{j}\cdot(3\mathbf{i} + \mathbf{k}) = 6\mathbf{i}\cdot\mathbf{i} + 2\mathbf{i}\cdot\mathbf{k} - 3\mathbf{j}\cdot\mathbf{i} - \mathbf{j}\cdot\mathbf{k}$
$= 6 + 0 - 0 - 0 = 6$

If $A = 2i + 3j - 4k$ and $B = 3i - j + 2k$

find (i) $A \cdot B$

(ii) Magnitude of $|A|$ and $|B|$

(iii) Angle θ between Vectors A and B

Solution

Given that

i) $A = 2i+3j-4k$ and $B = 3i-j+2k$

$$A \cdot B = (2i+3j-4k)\cdot(3i-j+2k)$$
$$= (2\times3)i\cdot i + (3\times-1)j\cdot j + (-4\times2)k\cdot k$$
$$= 6 + (-3) + (-8)$$
$$= -5$$

ii) $|A| = \sqrt{i^2+j^2+k^2} = \sqrt{2^2+3^2-4^2} = \sqrt{4+9+16} = \sqrt{29}$

$|B| = \sqrt{3^2-1^2+2^2} = \sqrt{9+1+4} = \sqrt{14}$

iii) $A\cdot B = |A||B|\cos\theta$

$$\cos\theta = \frac{A\cdot B}{|A||B|} = \frac{-5}{\sqrt{29}\sqrt{14}} = -0.248$$

$$\theta = \cos^{-1}\left(\frac{A\cdot B}{|A||B|}\right)$$
$$\theta = \cos^{-1}(-0.248) = 104.359^\circ$$
$$\theta = 104.36^\circ$$

CROSS OR VECTOR PRODUCT

The cross or vector product of **A** and **B** is a vector $\mathbf{C} = \mathbf{A}\times\mathbf{B}$ (read **A** cross **B**). The magnitude of $\mathbf{A}\times\mathbf{B}$ is defined as the product of the magnitudes of **A** and **B** and the sine of the angle between them. The direction of the vector $\mathbf{C} = \mathbf{A}\times\mathbf{B}$ is perpendicular to the plane of **A** and **B** and such that **A**, **B** and **C** form a right-handed system. In symbols,

$$\mathbf{A}\times\mathbf{B} = AB\sin\theta\,\mathbf{u}, \qquad 0 \leqq \theta \leqq \pi$$

where **u** is a unit vector indicating the direction of $\mathbf{A}\times\mathbf{B}$. If $\mathbf{A}=\mathbf{B}$ or if **A** is parallel to **B**, then $\sin\theta = 0$ and we define $\mathbf{A}\times\mathbf{B} = 0$.

The following laws are valid:

1. $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$ (Commutative Law for Cross Products Fails)
2. $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$ Distributive Law
3. $p(\mathbf{A} \times \mathbf{B}) = (p\mathbf{A}) \times \mathbf{B} = \mathbf{A} \times (p\mathbf{B}) = (\mathbf{A} \times \mathbf{B})p$, where p is a scalar.
4. $\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = 0$, $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$
5. If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, then

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}$$

6. $|\mathbf{A} \times \mathbf{B}|$ = the area of a parallelogram with sides **A** and **B**.
7. If $\mathbf{A} \times \mathbf{B} = 0$ and **A** and **B** are not null vectors, then **A** and **B** are parallel.

PARALLELOGRAM LAW OF VECTORS

The parallelogram law of vectors is another useful law for the addition of two vectors. Consider two vectors **P** and **Q** as shown in Figure 1.9a. Displace vector **Q** parallel to itself till the tail of **Q** touches the tail of **P**. Complete the parallelogram as shown in Figure 1.9b. Applying the triangle law of vectors to the vector triangle OAC, we get

$$\mathbf{OA} + \mathbf{AC} = \mathbf{OC} \quad \text{or} \quad \mathbf{P} + \mathbf{Q} = \mathbf{R}$$

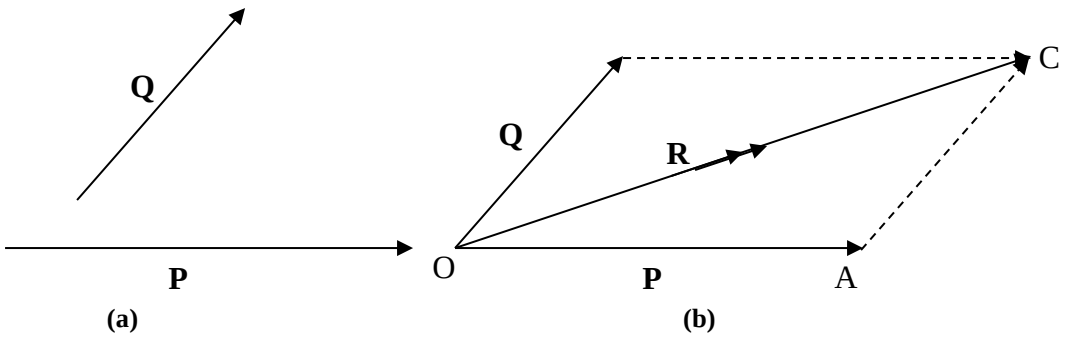


Figure 1.9: (a) P are **Q** two separate vectors **(b)** vector **Q** is displaced parallel to itself till its tail touches the tail of **P**. Broken lines are then constructed to complete the parallelogram.

So, the two vectors are represented completely by the two adjacent sides of the parallelogram and the diagonal drawn through the meeting point of the tails gives the resultant vector.

The parallelogram law of vectors states that if two vectors, acting simultaneously at a point, can be represented both in magnitude and direction by the two adjacent sides of a parallelogram drawn from a point, then the resultant is represented completely both in magnitude and direction by the diagonal of the parallelogram passing through that point.

Now, let us analytically calculate the magnitude and direction of the resultant vector R.

Let θ be the angle between the two given vectors **P** and **Q**. Refer to Figure 1.10.

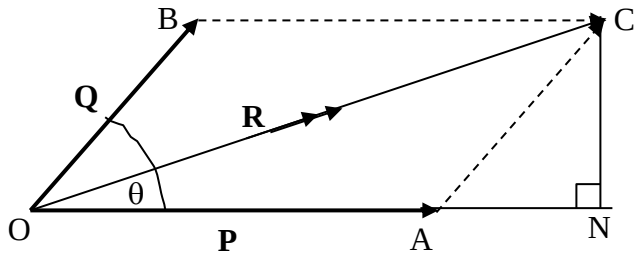


Figure 1.10:

From C we draw a perpendicular CN on an extension of OA to form a right-angled triangle ANC so that

$$\sin\theta = \frac{CN}{AC} \quad \text{or} \quad CN = AC\sin\theta \quad \text{or} \quad CN = Q\sin\theta\ , \qquad \textbf{1.14}$$

$$AC = OB = Q \qquad \textbf{1.15}$$

and also,

$$\cos\theta = \frac{AN}{AC} \quad \text{or} \quad AN = AC\cos\theta \quad \text{or} \quad AN = Q\cos\theta \qquad \textbf{1.16}$$

Now, $ON = OA + AN = P + Q\cos\theta$

Consider the right-angled triangle ONC.

$$OC^2 = ON^2 + CN^2 \qquad \textbf{1.17}$$

Or

$$R^2 = \left(P + Q\cos\theta\right)^2 + \left(Q\sin\theta\right)^2$$

$$R^2 = P^2 + Q^2\cos^2\theta + 2PQ\cos\theta + Q^2\sin^2\theta$$

$$R^2 = P^2 + Q^2\cos^2\theta + Q^2\sin^2\theta + 2PQ\cos\theta$$

$$R^2 = P^2 + Q^2\left(\cos^2\theta + \sin^2\right) + 2PQ\cos\theta$$

$R^2 = P^2 + Q^2 + 2PQ\cos\theta$ since $\cos^2\theta + \sin^2\theta = 1$. Then

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\theta} \qquad \textbf{1.18}$$

Equation 1.18 is the required expression for the magnitude of the resultant of two vectors **P** and **Q** inclined to each other at an angle θ.

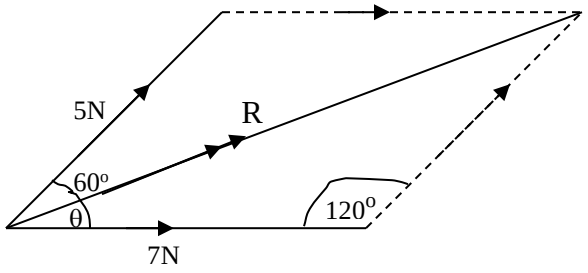
Example 1.8. Two forces 5N and 7N act on a 10kg body. If the angle between the two forces is 60°, calculate the magnitude and direction of the resultant force on the body.

Solution

Completing the parallelogram gives R as the resultant force acting on the body.

Applying the cosine rule,

$R^2 = 5^2 + 7^2 + 2(5)(7)\cos 60^o$



$R = \sqrt{25 + 49 + (0.5)70} = 10.44N$

Using the sine rule to find the direction of the resultant force gives

$\frac{\sin\theta}{5} = \frac{\sin 120}{10.44} \Rightarrow \sin\theta = \frac{5}{10.44}\sin 120$

$\theta = \sin^{-1}(0.415) = 24.5^o$

Trial questions

1.1 Show that the expression $V=V_o + at$ is dimensionally correct, where V and V_o represent velocities, a is acceleration, and t is a time interval

1.2 Suppose we are told that the acceleration **a** of a particle moving with uniform speed V in a circle of radius r is proportional to some power of r, and some power of v. Use dimensional analysis to predict the form of the relation. **(Ans: $a = Kr^{-1}V^2 = KV^2/r$)**

1.3 If $x = a + bt + ct^2$ where x is in metres (m) and t in seconds (s), then what is the unit of c? **(Ans: ms^{-2})**

1.4 The frequency of vibration, f, of a mass m, at the end of a spring that has stiffness constant k is related to m and k by the relation of the form $F = (\text{constant})m^a k^b$. Use dimensional analysis to find a and b. It is known that $[F] = [T]^{-1}$ and $k = [M][T]^{-2}$ **(Ans: a= -1/2 ; b= 1/2)**

1.5 Use dimensional analysis to determine which of the following equations is certainly wrong: $\lambda = vt$, $F = m/a$, $F = mv/t$, $h = v^2/2g$, $V = (2gh)^{1/2}$ **(Ans: F=m/a)**

1.6 The micrometer (1μm) is often called the micron. How many microns make up 1.0km? **(Ans: 10^9 μm)**

1.7 Express the speed of light 3.0×10^8 ms^{-1} in millimeters per picosecond (mm/ps). **(Ans: 0.3mm/ps)**

1.8 A force of 3N acts at 90° to a force of 4N. Find the magnitude and direction of their resultant, R**. (Ans: R=5N, θ=37 °)**

1.9 A particle which is moving due east at 4ms^{-1} changes direction and starts to move due south at 3ms^{-1}. Find the change in velocity. **(Ans: velocity = 5ms^{-1} at 37 ° S of W)**

1.10 Calculate the horizontal and vertical components of a force of 50N which is acting at 40 ° to the horizontal. **(Ans: Horizontal = 38N ; Vertical = 32N)**

Use the information below to answer questions 1.11-1.13. Consider two vectors $\mathbf{A} = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{B} = -\mathbf{i} - 4\mathbf{j}$. Calculate:

1.11 (i) $\mathbf{A} + \mathbf{B}$; (ii) $\mathbf{A} - \mathbf{B}$. **(Ans: [i] $2\underline{i} - 6\underline{j}$, [ii] $4\underline{i} + 2\underline{j}$)**

1.12 (i) $|\mathbf{A} + \mathbf{B}|$; (ii) $|\mathbf{A} - \mathbf{B}|$. **(Ans: [i] $\sqrt{40}$ units, [ii] $\sqrt{20}$ units)**

1.13 The direction of (i) $|\mathbf{A} + \mathbf{B}|$; (ii) $|\mathbf{A} - \mathbf{B}|$ **(Ans: [i] $\theta = 288^{o}$, [ii] $\theta = 26.57^{o}$)**

1.14 The resultant force on a body of mass 4.0kg is 20N. What is the acceleration of the body? **(Ans: 5.0ms^{-2})**

1.15 A body of mass 3.0kg slides down a plane which is inclined at 30° to the horizontal. Find the acceleration of the body: (g=10ms^{-2}) (a) if the plane is smooth (b) if there is a frictional resistance of 9.0N. **(Ans: [a] 5.0ms^{-2}, [b] 2.0ms^{-2})**

1.16 A body of mass 6.0kg moves under the influence of two oppositely directed forces whose magnitudes are 60N and 18N. Find the magnitude and direction of the acceleration of the body. **(Ans: 7.0ms^{-2} in the direction of 60N force)**

1.17 A body of mass 5.0kg is pulled along smooth horizontal ground by means of a force of 40N acting at 60° above the horizontal. Find: (a) the acceleration of the body (b) the force the body exerts on the ground. (g = 10ms^{-2}) **(Ans: [a] 4ms^{-2}, [b] 15.4N)**